IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 13 May 2023, accepted 16 May 2023, date of publication 22 May 2023, date of current version 1 June 2023.

Digital Object Identifier 10.1109/ACCESS.2023.3278267

RESEARCH ARTICLE

Digital Twin for Safety and Security: Perspectives on Building Lifecycle

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This work was supported by the Direct Operations Project under Grant 323831.

ABSTRACT As digital twins are gaining importance in various industries, research on their impact, use cases, and implementation in the field of construction is accelerating. In this study, we examine the impact of digital twins on building safety and security throughout their lifecycle, utilizing two state-of-the-art use cases for fire and anomaly detection. The solution employs advanced sensors, 3D visualizations for detecting safety and security threats, and data analysis to predict potential future threats. Using the data that was collected over a seventeen-month period, we present an implementation architecture for safety and security digital twins in buildings, explore the benefits and shortcomings of such a system, and discuss the learnings applicable to different lifecycle stages of a building. This paper contributes to the creation and study of a digital twin for fire safety, analyzing its performance using two real-world use cases and reporting the implementation results. Additionally, we extrapolate the learnings from the use phase of a facility to other lifecycle phases of buildings and facilities.

INDEX TERMS Digital twin, building lifecycle management, fire safety, safety and security in buildings.

I. INTRODUCTION

The building safety and security is a crucial topic in today's digitalized world, which encompasses elements such as fire safety, environmental (heating, ventilation, and air conditioning) control, and surveillance and access monitoring. Safety refers to the intention of safeguarding from the impacts of accidents on health of people [1], and security means protecting assets such as information and resources from intentional attacks [2]. Safety and security requirements are essential throughout the building lifecycle. For instance, according to US Fire Administration, in 2020, 3500 people lost their lives to fire related accidents and there were 15200 injuries caused by fire, only in the United States [3]. Moreover, losses caused by fire incidents, in 2020, in the US stood at $21.9 billion [3]. With respect to building security, in the US, there were an estimate of 1,117,696 burglaries in the year 2019 and the victims of theft offenses suffered an estimated $3.0 billion in property losses in 2019 [4]. Thus, the size and the importance of the losses and damages caused by safety and security shortcomings in the residential and commercial buildings are vast [5] and improvements in these areas may significantly reduce human loss and injuries [6], and financial burden caused by the incidents [7]. Using digital twin (DT) for safety and security has the potential to enhance the current situation by offering betterments in detecting fires as well as unauthorized access to the buildings, and building's subsystems malfunctions [8].

Smart buildings are an emerging phenomenon that refers to buildings integrating and accounting for intelligence, materials, construction, and control as a cohesive system with adaptability, efficiency, longevity, and comfort in mind [9]. There is a need for interactivity and responding to dynamic changes in inhabitants' needs in the buildings [10]. It is possible to adjust lighting, heating, cooling and provide better security as well as data on how the building has been

The associate editor coordinating the review of this manuscript and approving it for publication was Hang Shen.

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used [11]. The distinction between the conventional buildings and the smart buildings is related to the availability of the IoT (Internet of Things) systems, cloud-based services, and analytics measures. For instance, IoT enabled control systems in smart buildings can improve the lighting power consumption and enhance the quality of lighting control operations [12]. Hence, smart buildings have the possibility to bring together the technologies such as IoT sensors and collected data, to better understand and analyze the processes within a building [13]. Moreover, smart alarm systems in smart buildings can signal unexpected events to help inhabitants evacuate or take reactive measures [14]. In the context of smart buildings, an emerging stream of research investigates the use of digital twins for real-time and agile monitoring, simulation, and data-fusion for further improvements in various building services [15], [16]. A building digital twin is a dynamic and digital representation of a building that can depict the past, present and simulated futures of its physical twin [17]. The concept of digital twin for safety is becoming increasingly important [18]. There is high potential in the construction industry for finding solutions to various problems through digital twin applications [19]. However, there remain various challenges to be addressed regarding applicable and feasible use cases, implementation approaches, and prohibitive technical shortcomings.

In this research, we explore how a digital twin can be implemented and improve the safety and security of a building throughout its lifecycle, and effectively detect unexpected events in the buildings by utilizing real-world case studies in fire and anomaly detection. There is a gap in research and literature on how digital twins can help improve the safety [20] and security [21] of buildings and what the lifecycle implications are. We intend to address this gap by answering the following three research questions:

1) How to implement digital twins for safety and security of buildings throughout their entire lifecycle?
2) What are the real-world benefits that can be gained from the implementation of digital twins for safety and security in commercial and residential buildings?
3) What challenges and limitations exit in the development and operation of safety and security digital twins for buildings?

The contribution of this paper lies in the development and detailed examination of a digital twin for building safety, utilizing two real-world use cases. We analyze its performance, report on the implementation, and discussed the results derived from both use cases.

The remaining sections of the paper are organized as follows; After the Introduction, we present the Literature Review, in which we examine four research streams: the building lifecycle, smart buildings, digital twin and its applications in building, and safety and security in buildings context. Subsequently, we describe the research Methodology, followed by the Findings section presenting our results. We then engage in a Discussion and conclude the paper with the Conclusions section.

II. LITERATURE REVIEW

This research combines four essential research streams within the context of modern building management: the building lifecycle, smart buildings, digital twin and its applications in building, as well as safety and security in buildings context. By examining these interconnected domains, we provide a thorough understanding of the current state of research.

A. THE BUILDING LIFECYCLE

A lifecycle, generally, refers to the series of stages or phases that an object, system, or process undergoes from its inception to its conclusion or eventual end. The lifecycle of a building can be divided into four main stages which are design, construction, exploitation, and maintenance stages [22]. According to CEN/TC 350 which is the European Committee for Standardization's Technical Committee 350 for construction works, a building's lifecycle is divided into four stages [23]:

1) Product phase which refers to producing and supplying the raw materials to the site.
2) Construction phase that refers to the on-site operations and processes such as installing materials on-site.
3) Use phase which means maintenance and repairs, operational use such as heating, cooling, ventilation, lighting as well as water use.
4) End-of-life phase, which is deconstruction, transport, recycling and reuse as well as discarding

By integrating the two building lifecycle breakdowns [22], [23] and to offer a more comprehensive analysis in this research, we propose a building's lifecycle consisting of predesign, design and construction planning, site preparation, construction, use during the occupancy, and maintenance, renovation, and demolition phases, as illustrated in Fig 1.

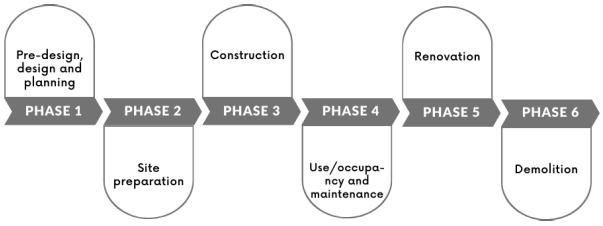


FIGURE 1. Six phases of a building lifecycle.

1) SOLUTIONS TO MANAGE LIFECYCLE

Building Lifecycle Management (BLM) brings together multidisciplinary teams involved throughout the building lifecycle and integrates the various processes associated with a building's lifecycle [24]. This holistic approach ensures that all stages of a building's life are taken into account and managed efficiently. One of the solutions widely utilized in the construction industry for managing the lifecycle of facilities is Building Information Modelling (BIM). BIM is a system for storing the building information in an accurate and interoperable way to improve planning, construction and maintenance of a building over its lifecycle [25], [26].

Although BIM is not a real-time system, it provides a digital historical representation of the functional and physical characteristics of a facility, serving as a shared information source that can aid in decision-making processes [27]. BIM has been developed to embed building-related data, such as schedule, cost estimation, maintenance, and other specifications, into a digital 3D representation of the building model [28], [29]. However, BIM was not designed to be real-time and, in most cases, needs to be updated manually after major renovations or changes to the building in order to accurately represent the current state of the facility [30].

BIM is highly beneficial for safety and security [31], including construction safety planning, maintenance safety planning, emergency planning and management, occupational safety planning, safety analysis, and fire engineering [32], [33], [34]. In terms of security, BIM can aid in security design, such as in conduit and conductor routing and sizing, device placement for intrusion detection and access control, and CCTV, as well as performing automated fire safety code reviews [31]. By linking RFID tracking data with BIM, managers can monitor the flow of people in and out of different zones, however, the system cannot pinpoint the exact location of a worker [35]. An automated system has been proposed that reacts to alarms by retrieving alarms reported by facility management systems for corrective maintenance, with the data written into Industry Foundation Classes (IFC) BIM file when needed [36].

However, despite the advantages and possible use cases for building operations, BIM is still not able to solve all challenges faced during a building's lifecycle. For instance, the digital design models are often not kept up to date throughout the building lifecycle [30]. Although a BIM model may contain all the necessary information to construct and operate a building, it does not have an explicit connection to the physical entity [37]. Implementing BIM requires a multifaceted approach involving multiple stages and is influenced by project networks and supply chain interdependencies [38]. Furthermore, over 80% of buildings in Europe do not have BIM models [25].

B. SAFETY AND SECURITY IN BUILDINGS CONTEXT

Safety and security are essential for all buildings, both during construction and operation, as they help minimize the risk of injuries, accidents, and crime. Safety refers to the absence of unacceptable risk of physical injury or harm to people's health [39]. As such, building safety and security aim to protect the inhabitants and the building itself from dangerous situations and events that may cause harm or damage.

To safeguard the building and its occupants, it is vital to equip buildings with appropriate security systems and processes. This not only helps to maintain operational capacity but also minimizes the potential for loss of life and financial damage. Adherence to modern safety and security standards ensures the protection of both the building and its inhabitants while preserving its operational capabilities [40].

TABLE 1. Safety and security components at building's lifecycle phases.

LCP* \ SaS Systems	AC	HVAC	LC	FAS	VS	FM
Design and planning	+					
Site preparation	+		+	+	+	
Construction	+	+	+	+	+	
Use and maintenance	+	+	+	+	+	+
Renovation	+	+	+	+	+	
Demolition	+				+	

*The abbreviation of the table header: LCP stands for lifecycle phases; SaS systems stands for safety and security components; AC stands for access control [35], [45]; LC stands for lighting control [44]; HVAC stands for heating, ventilation, and air conditioning [46]; FAS stands for fire alarm systems [45]; VS stands for video surveillance [44]; FM stands for facility management [47]

1) CURRENT SAFETY AND SECURITY SYSTEMS IN BUILDINGS

Various systems and technologies have been utilized to provide safety and security for buildings during both the construction and use phases. These include building automation systems (BAS), fire protection systems, natural disaster protection systems, access control systems, intrusion detection systems, indoor air quality systems, energy information management, and lighting control systems.

During the construction phase of a building, Bluetooth Low Energy (BLE) beacons can be used to monitor the harness use of workers [41]. Additionally, in the use phase, BLE can be integrated into a mobile solution for on-premise security monitoring using Android devices and Bluetooth Low Energy beacons. This system incorporates two mobile-based applications, beacons, and supporting cloud services [42]. However, a shortcoming of BLE applications is that the signal coverage is affected by the placement of the devices, and walls can impact the accuracy [43]. The Table 1 presents the safety and security components [44] visible in each building lifecycle phase.

During the use phase, various systems such as building automation and control systems (BACS), home automation systems (HAS) or building automation systems (BAS), fire protection systems, natural disaster protection systems, access control systems, video surveillance, and intrusion detection are employed. BACS consist of products and services designed to automatically control, monitor, optimize, operate, engage individually, and manage, ensuring energy-efficient, cost-effective, and safe/secure operation of the building [1], [48].

For instance, home automation systems (HAS, referring to small buildings, particularly in residential areas) and building automation systems (BAS, used in large buildings such as office buildings, hotels, and hospitals) are smart building technologies that control and monitor various systems. These systems include lighting (with presence detection and crepuscular outside light control), heating/cooling, ventilation and air conditioning (HVAC), energy management (household appliances and other electric devices with energy

smart metering), security and access control systems (anti-burglary systems), remote surveillance systems, and other functions. Additional features consist of window blinds, communication equipment (intercom systems, telephones, public address audio systems), elevators, indoor and outdoor irrigation (with sensors and weather-station feedback), water supply systems, audio/video controls (on/off, volume, etc.), gutters, stairs, alleys and terraces de-icing systems, and garage and exterior gate control [49].

Fire protection systems in buildings are designed to detect and suppress fires, as well as guide occupants to safety in the event of a fire. The main components of a fire protection system include fire detection and alarms (using heat or smoke sensors), fire suppression systems to extinguish the fire (such as sprinkler systems and gas suppression systems), fireproofing with the use of fire-resistant construction materials and techniques to slow the spread of fire, emergency lighting and exit signs, and firefighters' communication systems. There are smoke detection systems [50], [51] for preventing casualties in fire-related incidents; however, smoke detectors have a tendency for false alarms, and the detection of fire through this method may not provide accurate location information.

Regardless of a building's smart features, independent fire alarm systems that are not connected to a network are used in residential buildings. In the context of commercial buildings, each building has its own control system and is connected to the fire department. Fire alarm systems are considered complicated, costly, and uncompact [52], and are triggered only after fire incidents, meaning they cannot monitor temperature fluctuations in real-time [53].

Another essential system for safety and security in buildings is the access control system. This system regulates entry by utilizing components like card readers, sensors, a central host server, and a controller [44]. The HVAC system, which includes boilers, chillers, and air terminal units, maintains desired indoor environmental conditions. Lighting control is achieved using sensors, timers, and control units. Additionally, video surveillance involves installing cameras that are remotely controlled by a dedicated security control center [44].

C. SMART BUILDINGS

Smart buildings utilize technological systems to improve system functionality, achieve better financial results from integrated systems, and realize energy savings [54]. Smart buildings monitor the living environment and sense the actions of their inhabitants, observing occupancy and behavioral patterns while also attempting to predict future states of the living space [55]. This allows for autonomous operations through continuous sensing and responding, enabling the system to integrate dynamic user information and optimize the building's operations [56].

The architecture of a smart building consists of sensors, a centralized control system, network connections, controllers, and an interface to communicate with the network [57]. The distinction between older buildings and smart buildings is related to the availability of sensor systems [58] and analytics measures [59]. Smart environments can create safer living environments and have the potential to impact inhabitants' lives in many ways, such as reducing human encounters in a building during a pandemic [60]. A smart system can intelligently respond to future events and monitor various aspects of a building, including energy usage, water consumption, safety, and security [61], [62]. Smart building technologies can integrate with various building management systems like Building Management System (BMS), Fire Alarm System (FAS), Building Automation and Control Systems (BACS) which make them more efficient [63]. Smart buildings also may incorporate elements of sustainable design, such as solar panels or green roofs, to reduce their environmental impact.

An area of smart building research is focused on the use of data and artificial intelligence (AI) for analytics to improve the performance of buildings [64], [65]. Smart building research is becoming increasingly interdisciplinary, with researchers from various fields such as computer science, electrical engineering, architecture, and environmental science, working together to develop new solutions to address the complex challenges faced by building owners and operators.

D. DIGITAL TWIN AND ITS APPLICATIONS IN BUILDING

The concept of digital twin has been widely studied in engineering context, particularly for creating real-time digital models of physical systems such as jet engines. However, research on the implementation and applications of digital twins in the construction industry has only recently gained momentum [15], [66], [67]. By creating a digital model of a building that receives continuous real-time data from various sensor networks installed in the building, multiple aspects of the building can be improved. These improvements include enhancing the comfort and well-being of its occupants, improving the energy efficiency of subsystems such as lighting and HVAC, and increasing the overall sustainability of the building [15].

The novelty of building digital twins lies in their ability to provide visibility into the system and offer additional services, such as predictive maintenance, through real-time data exchange between physical and digital assets [68]. Previous research has identified three main components of a building digital twin: sensor networks, data analytics, and 3D visualization [15]. Another study emphasizes the importance of data exchange between the physical and digital worlds, focusing on semantic and real-time data, as well as data processing that includes communication protocols allowing the exchange between the physical and digital realms [69].

The integration of digital twins with Building Information Modeling (BIM) can lead to improvements in asset maintenance, asset tracking, and equipment failure prediction [70]. Furthermore, digital twins can enhance safety management methods and modes of construction operations. A digital twin system that combines IoT, BIM, and security risk analysis

can be applied to prefabricated building hoisting safety management, thereby improving the data-driven mechanism of hoisting safety risk management frameworks [71]. However, by dividing the timeline of an incident into three stages: pre-, during, and post-incident, we can differentiate between BIM and digital twin in a building context. Unlike BIM, which mainly contributes to pre-incident preparation, the real-time availability and functionality of a digital twin make it valuable and crucial also during and after incidents [15].

The concept of a digital twin of a building is related to the smart building, but it is distinctive. A digital twin of a building is a digital representation of a building [15], while a smart building is a physical residential or commercial building that utilizes technologies to automate and improve its performance [61]. One of the main differences between smart building systems and building digital twins is the emphasis on the use of 3D models in digital twins. Data from many different sensors can be integrated into a dimensionally accurate 3D model of the environment in real-time. The model allows for real-time integration and visualization of building conditions while also providing simulation possibilities for training, predictive maintenance, and other advanced analysis purposes using multi-model optimization and scenario modelling.

In contrast, smart building systems are primarily focused on optimizing building performance, energy efficiency, and occupant comfort through the integration of sensors, IoT devices, and automated control systems. In summary, digital twin of a building can provide actionable data for smart building systems. Moreover, digital twin of a building allows for simulation, testing and validation of changes in the digital model of the building before real-world implementation, which can lead to more efficient and cost-effective building lifecycle management.

E. GAP IN THE LITERATURE

There is a gap in the literature on how digital twin technology can improve the safety [20] and security [21] of buildings, as well as the implementation and performance of a real-world digital twin for safety and security application throughout a building's entire lifecycle [72]. Previously, an initial BIM-based digital twin framework has been outlined [20]. Camera-based fire detection systems have been proposed [53]. A digital twin security simulation was made as a process-based security framework [21]. However, there has not been any research illustrating how the digital twins can help improve building safety and security with real-world applications. We contribute to filling of the literature gap related to digital twin application areas in the building lifecycle.

III. METHODS AND MATERIALS

Application of digital twins is a novel approach and the real-world implementations and use cases for safety and security in the context of buildings are very scarce. We used Eisenhardt's [73] case study recommendations since, case study method is especially suitable for revelatory cases where an observer have access to a phenomenon which was previously inaccessible [74]. This research is based on two real-world case studies of implementing digital twin for safety in building environment. A digital twin solution for detecting the fire and other anomalies causing fires and safety related incident is developed and installed in two different indoor spaces to explore the architecture, benefits, capabilities, and shortcomings of a building digital twin for safety and security. The case studies investigated for this research are:

1) First case study: Fire monitoring inside an office space,
2) Second case study: Anomaly detection in a grain drying facility.

The data was collected between August 2021 and December 2022 in Finland. The digital twin solution used in both case studies consisted of the following components which can also be seen in Fig 2:

- A customizable sensor package consisting of various sensors.
- The 3D visualization of the environment to integrate and present the real-time status of the environment.
- The backbone and data analytics component comprises an application programming interface (API) that gathers data from the sensors, enabling sharing and analysis of the collected information for presentation to the client via the web.

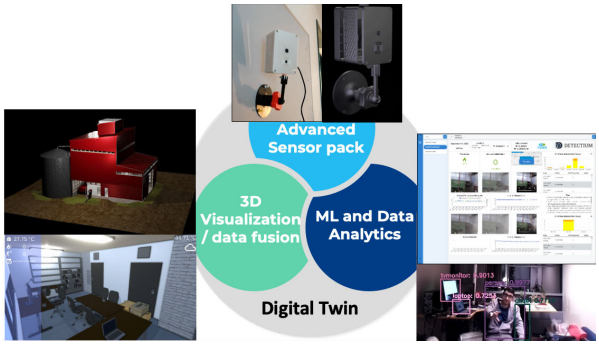


FIGURE 2. The components of a digital twin for safety.

The sensor packages utilized a Raspberry Pi 4 single-board computer for in-situ computing and a variety of sensor types. In both cases, the 3D model was created in Blender software, implemented in Unity and rendered in WebGL. In both cases, REST API was employed to upload the data collected from the sensors to Microsoft Azure cloud where the dashboard was implemented.

A. SENSOR PACKAGES

The first case study made use of the sensor package depicted in Fig 3 (a) and (b), which included a motion detector, a short-range thermal camera, an RGB camera, and a UV sensor. The UV sensor, specifically designed for long-range fire detection, was a prototype under development.

In the second case study, the sensor package was modified. While an RGB camera was still included, a different thermal

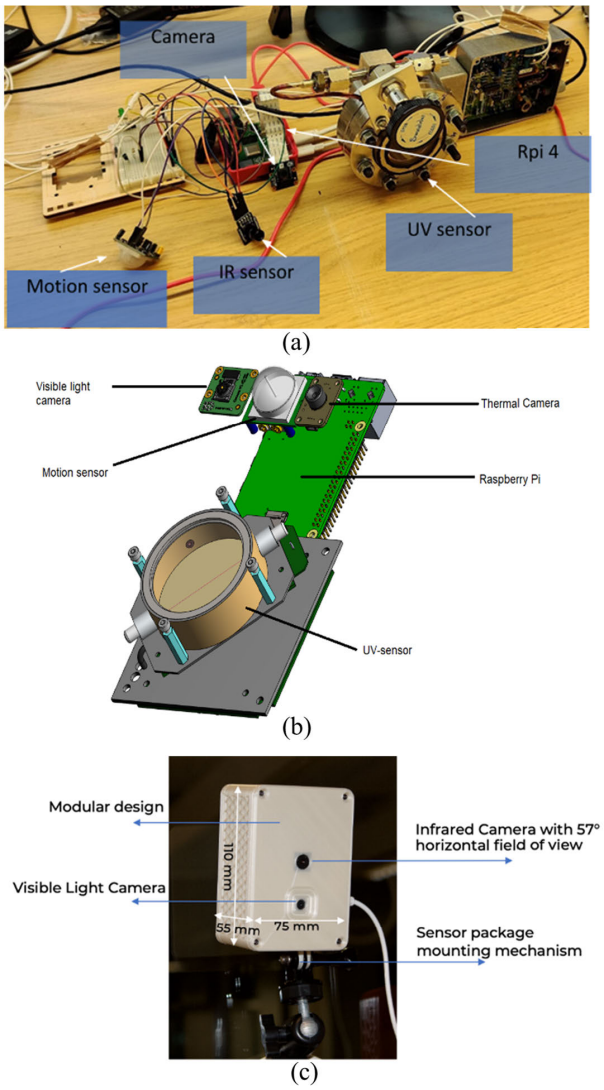


FIGURE 3. (a) The sensor package utilized in the first case study, including the UV sensor prototype in the lab. (b) 3D design rendering of the sensor package. (c) Sensor package used in second case study.

sensor with a longer range was utilized. The motion detector was removed, and an internal temperature sensor was added instead. Fig 3 (c) illustrates the sensor package employed in the second case study.

B. REAL-TIME VISUALIZATION AND SENSOR FUSION

For real-time visualization, we manually created a dimensionally accurate 3D model of the buildings and spaces using LiDAR scanners and point clouds, or by referencing building design plans. We then implemented sensor data fusion using Unity software and WebGL. This approach enabled us to display changes in fire and anomaly situations in real-time within the 3D model, which was stored on an Azure-based cloud dashboard.

C. DATA ANALYTICS

In the first case study, the sensor package monitored changes in UV emissions in the environment. Since fire flames emit UV, UV detection was chosen for this purpose. The system was designed to detect flames, rather than smoke or heat. If the UV sensor detected a flame, the RGB and thermal cameras would activate, recording a short video clip and uploading it to the cloud for the building manager or system operator to verify the presence of a fire. Consequently, data collection was triggered by the UV sensor, rather than the RGB or infrared sensors.

In the second case study, we relied on the thermal camera for anomaly detection. The thermal camera continuously monitored the space, and the code detected the presence of high or increasing temperatures beyond a specified threshold. When this occurred, the RGB camera would record a short video and take a photograph, uploading both to the cloud dashboard. Again, the building manager or system operator would determine the accuracy of the detection. In the second case study, which focused on a grain drying facility, we used two sensors, while in the first case study, which took place in an office space, we used only one sensor.

In both case studies, we extracted and analyzed data from the database on an incident-by-incident basis during and after the testing period. In the first case study, this allowed us to analyze false detections as well as real fire hazards detected by the sensors, differentiating them from the existing fire detection solution, which was a smoke detector in the office. In the second case study, there wasn't a fire detection solution, as regulations do not require farms to have a fire detection system. Consequently, our system served as the primary fire detection system.

In the first case study, it was possible to compare the two fire detection systems. The comparison logic was straightforward; by conducting tests with multiple fire sizes, we could examine the sensitivity of both the conventional and the new digital twin-based system. In both case studies, data was collected during the building use phase. However, the learnings will be discussed in terms of extrapolation throughout the building's lifecycle.

IV. FINDINGS

In this section, we present the key findings of the paper to answer the research questions. The contribution of this paper lies in the development and examination of a building digital twin for safety through the implementation of two real-world case studies. We analyze the performance of the digital twin and report the outcomes and insights gained from these case studies. In this work, we also extrapolated the insights gained from the use phase of facilities to the other lifecycle phases of buildings and facilities.

A. SAFETY DIGITAL TWIN IMPLEMENTATION ARCHITECTURE FOR BUILDINGS

In this section, we study the implementation of a digital twin for fire safety in a building during its use phase.

Based on sensor observations (e.g., infrared and visible light) of the monitored environment or system over a predetermined time period, a normal operational signature (NOS)

status is created and saved for the system. The infrared NOS, while both sensors and the system operate in a stationary mode, is determined by establishing lower and upper temperature boundaries for each pixel in the infrared sensor image, as well as the rate of temperature change for each pixel at any given time.

The NOS is then used to detect anomalies in the machinery or environment. The analysis system continuously and in real-time compares the sensor input results with the NOS, searching for pixels with significant deviations. When the out-of-NOS situation persists for a predetermined period of time, secondary notifications and system status updates are performed on the real-time 3D environmental model. For instance, an out-of-NOS condition below the established signature can be interpreted by the system as a machine shutdown, while an out-of-NOS condition above the established signature can be interpreted as a machine anomaly caused by increased friction or other internal combustion issues.

The architecture of the digital twin for predictive anomaly detection, early fire hazard detection, and fire propagation simulation is depicted in Fig 4. The data collected from the sensors is locally analyzed, and the fire alarms are then uploaded to the backbone to be shared with the cloud-based analytics dashboard and the 3D model of the environment.

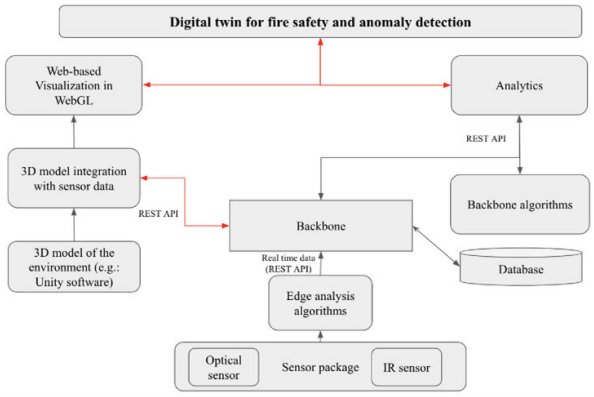


FIGURE 4. The architecture of the digital twin solution for safety and fire detection.

A dimensionally accurate 3D model of the environment is created for data fusion, with material types assigned to objects to assist with simulating fire propagation. The dimensionally accurate 3D model allows for the integration of data from multiple sensors and sources, improving the efficiency of monitoring. For instance, while the IR sensor and visible light camera are both direct line-of-sight sensors and cannot observe behind obstacles and walls, using multiple sensors and fusing the data in a single 3D visualized model helps resolve blind spot issues.

The use of a depth map derived from the dimensionally accurate 3D model of the environment enables the normalization of the IR sensor temperature measurements, thus alleviating inaccuracies caused by the impact of distance on the temperature values recorded by an IR sensor. To create a stereoscopic image of the scene, we can reposition the camera by an exact distance at the beginning of the sensor package installation, which allows us to create a depth map and subsequently assign depth as one of the criteria for each pixel. Later, when the system is in use, every temperature measured for a pixel can be normalized. The normalization process involves transferring pixels from different distance planes to a pre-specified plane using predetermined distance-dependent thermal value multipliers. This allows for accurate and distance-independent comparison of pixel temperatures, and temperature variation. In other words, while two objects with the same known temperature at different distances present different temperatures and temperature variations in raw IR sensor readings, normalization enables the temperature and its variation for both objects recorded by the IR sensor to be comparable. We propose two methods for depth detection: static and dynamic. The static method derives the depth map from the dimensionally accurate manually made 3D visualization model or from the stereoscopic image of the scene created by repositioning of the sensor package by an exact amount vertically or horizontally during installation and calibration, and using image analysis to calculate the depth map. Alternatively, a dynamic real-time measurement using LiDAR can be employed for more accurate normalization, as the movement of objects in the environment is not considered in a static depth map calculation based on images or a 3D model, unless the depth map creation process is repeated frequently.

1) FIRE LOCALIZATION WITHOUT TRIANGULATION

Using a single sensor package consisting of one infrared and visible light optical sensor, the location of any pixel can be synchronized, mapped, and displayed in the 3D environment model. This enables fire localization by accurately overlapping the two images taken from the same scene by the two aligned IR and visible light cameras and then finding the exact same pixels in the dimensionally accurate 3D model. In the first step, the fire detected in the IR image is mapped onto the visible light image, and the pixels corresponding to the fire are identified. In the second step, considering the exact location of the sensor package in the real world, the precise point of view of the visible light camera is determined in the dimensionally accurate 3D model. In the third step, the pixels are mapped and named based on the visible light camera's image. In the fourth step, the pixels identified in the first step are used to place the fire on the 2D image from the camera's point of view in the 3D model. Finally, using the assumption that fire always forms on floors and surfaces and not in mid-air, lines are drawn from the location of the camera lens's center in the 3D model to the direction of the fire pixels in the mapped image from step four. The points created from the first spots where the lines strike the 3D model surfaces in the direction of the fire pixels are marked as the fire location in the 3D model.

2) FIRE INTENSITY EVALUATION AND VISUALIZATION

By measuring and assigning depth values to the fire pixels, it is possible to estimate the actual size of the fire. During the fire localization process, determining the fire boundary pixels enables fire size estimation in the dimensionally accurate 3D model. Furthermore, in a detailed 3D model where material types are assigned to the scene's components, a real-time feed of the event in 3D can be provided to the fire brigade. Based on the size, shape, and location of furniture or equipment and their materials, a simple fire propagation simulation can be carried out in real-time, assisting the fire brigade in developing more accurate and effective fire extinguishment strategies.

3) ANOMALY DETECTION

The use of a digital twin for safety and fire detection also enables anomaly detection. While creating the 3D model for the digital twin, a heat signature for the system can be incorporated into the model, where the upper and lower temperature boundaries of a system can be determined. Different temperature ranges can even be assigned to various parts of a single machine when establishing the NOS. During the environment monitoring by a thermal camera, the system in real-time compares the current temperature of all system parts with the NOS from the digital twin. If temperatures outside of the NOS are detected, the digital twin can estimate the location of the anomaly in the 3D model, visualize it, and inform the operator. This can prevent fire events and mitigate damages that can cause long equipment downtime.

B. IMPLEMENTATION OF FIRST CASE STUDY

To create a digital twin of the office space for our first case study, we had to develop a 3D model. This 3D model was created in two incremental steps. First, we used the floorplan, layout of the office, and photographs of the location showing the arrangement of the furniture in the office space (Fig. 5 (a)) to add the walls and dimensions to the model using the ProBuilder feature of Unity software (Fig. 5 (b)). The development of the model was performed entirely in Unity, using ProBuilder. After the walls and floor of the space were created, we added the furniture and properties, such as desks, chairs, and shelves, to the model using the Unity software ''office Snaps'' package.

The first step of 3D visualization is critical, as it enables us to assign material types to the walls, machines, and furniture. This information can be utilized later for the analysis of fire propagation behavior and hazard mapping of the space.

In the second step, we added more details to the 3D model, such as the location of doors, windows, and lights. Fig 5 (c) displays the outcome of the second step.

Another component of our digital twin for safety was the cloud-based dashboard for historical data collection and analytics. This component consisted of an API that collects data from the sensors and enables sharing and analysis of the collected data for presentation to the client on the web. In this section, we briefly present the development of the

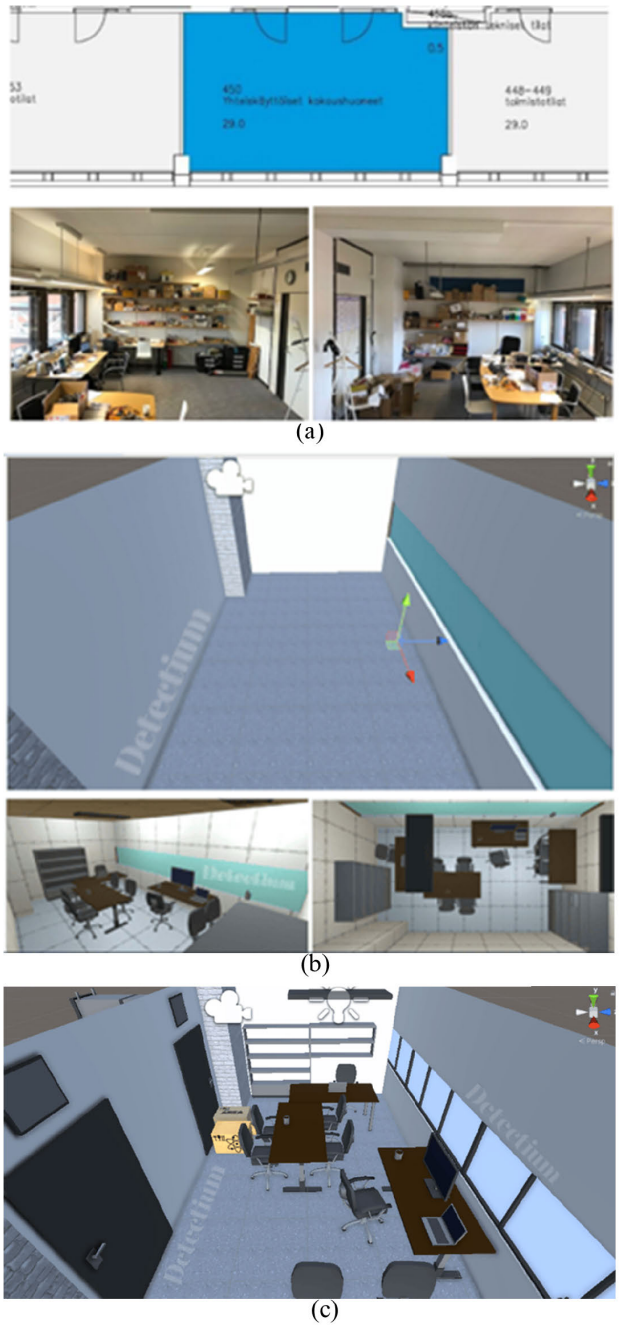


FIGURE 5. (a) The floor plan of the office and the corresponding images. (b) The first step in the 3D visualization of the office space in the Unity software. (c) The second step in the 3D visualization of the office space in the Unity software.

API, dashboard, and the backbone for data collection from the sensor package, along with the specifications of our solution concerning the numerical and graphical data. Furthermore, we discuss the logic behind the selection of analytics categories for the dashboard. Fig. 6 displays the login page, the dashboard visualization elements, as well as the 3D model with integrated sensor data on the cloud-based dashboard.

The cloud-based dashboard interface was developed to collect fire incident data in the environment, analyze it on the



FIGURE 6. The cloud-based dashboard features a login page, visualization elements, and a 3D model of the office space.

dashboard, and integrate it into the 3D model for simulation and visualization. A special module was built specifically for this project. The frontend platforms and technologies used for the development of the web interface include: Typescript (a JavaScript superset), Vue.js (a framework that implements the model-view-viewmodel pattern), HTML, and CSS.

The ''Analytics Dashboard'' containing visualization elements is one of the two pages accessible to users after successful login. In addition to sensor-collected data, other sources of third-party data, such as outdoor humidity, temperature, and weather forecasts, can be integrated into this page to enhance data analytics and meet customers' needs. The analysis dashboard visualization elements present statistical information and alerts on data received from sensors. These elements include the number of sensors, time and location of sensors, current status and alert of detected fire, images recorded of fire by the IR and visible light sensors, video recorded of fire by visible light sensors, fire alarms in the last seven days (histogram), fire hazard map of the location based on historical data, current and historical weather information of the location (e.g., temperature, humidity, etc.), sensor package internal temperature, and tips on fire accident prevention.

The second page accessible by users after login is the ''3D Model Visualization'' page. The real-time 3D visualization of the environment, including sensor data and fire status, is presented to the client on this page. This is integrated into the dashboard using WebGL. In the 3D visualization page, various types of information, such as the size and location of the fire, can be presented.

In case study one, a candle was placed 0.5 meters away from the sensor package. When the tester lit the candle, the digital twin quickly and accurately detected the fire in the office (in less than a second). The data was then uploaded to the digital twin's backbone using the API and presented in the 3D visualization model in real-time. The fire was accurately displayed in the 3D model on the table at the correct spot (Figure 7).

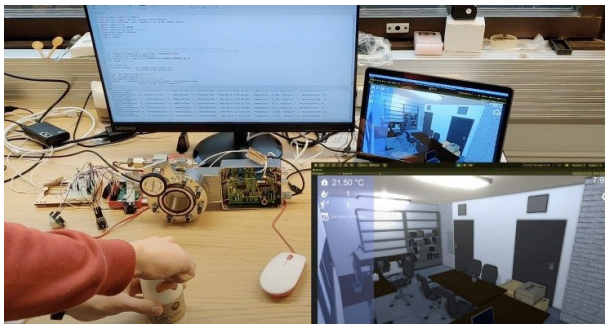


FIGURE 7. The candle-sized fire is detected and instantly displayed in the 3D visualization model of the digital twin.

The digital twin implemented for first case study was further improved to automatically mark the fire on the IR image, and to add a fire notification in the 3D visualization (Fig. 8). Additionally, the sensor package was further developed to allow for the production of a miniaturized UV sensor, suitable for a wall-mounted sensor package.



FIGURE 8. The enhanced and integrated view of the first case study digital twin 3D visualization, tested with mock data.

C. IMPLEMENTAION OF SECOND CASE STUDY

In the second case study, which focused on the deployment of two sensor packages at a farm, the sensors were placed at different locations within a grain drying building. The first sensor package (Senpack 1) was situated on the ground level (Figure 9 (a)) to monitor the hot air blower pipe (Fig. 9 (b)), which was not to exceed 80 degrees Celsius. Meanwhile, the second sensor package (Senpack 2) was installed on the third floor of the grain drying tower (Fig 9 (e)) to observe two electromotors and four ball bearings (Fig 9 (f)) that posed a fire risk. The facility's owner/operator set the recommended temperature limit for the bearings at 120 degrees Celsius.

A detailed 3D model of the farm was added to the cloud-based dashboard, which also included additional features such as picture-in-picture (PIP) for locations with sensors to ensure continuous monitoring of those areas. To create the digital twin, the detailed 3D model was integrated with real-time data from the sensors and third-party weather data. Additionally, it was implemented in Unity software with five camera points of view (Fig. 9 (c),(g)) and made available on the cloud-based dashboard using WebGL (Fig. 10 (a)).

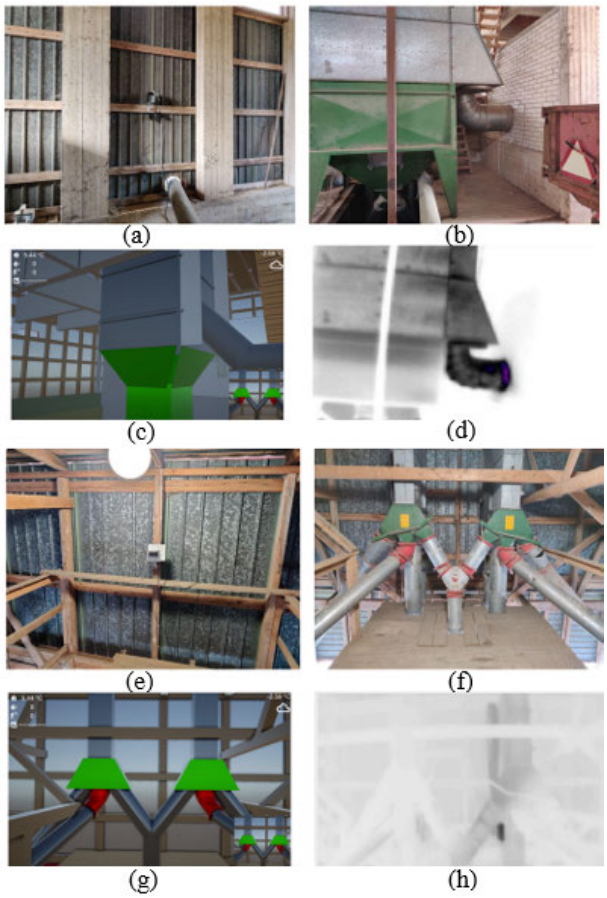


FIGURE 9. (a) Locations of Senpack 1. (b) Hot air blower pipe, point of view of the Senpack 1. (c) Senpack 1 point of view in digital twin 3D model. (d) The IR spectrum view from Senpack 1. (e) Locations of Senpack 2. (f) Two electromotors and four ball bearings, point of view of the Senpack 2. (g) Senpack 2 point of view in digital twin 3D model. (h) The IR spectrum view from Senpack 2.

Following the installation of Senpack 1 and 2 and the integration of their data into the 3D model (Fig. 10 (a)) and cloud-based dashboard (Fig. 10 (b)), we conducted tests with flames and confirmed that the digital twin was fully operational (Fig. 10 (c)). After three months of the sensors operating, we collected data to evaluate the digital twin's performance and analyzed the insights.

D. LOGIC BEHIND THE OPERATION OF SAFETY DIGITAL TWIN

The logic behind the fire and anomaly detection in safety digital twins used in both case studies is similar, with minor differences due to the use of different sensors for detection. In this section, we detail the operational process of the digital twin for safety and fire detection. Let f^t be a frame of the thermal image at time t, where each pixel in f^t contains a temperature value. h, w are the resolution of the thermal image, in our experiments, they are set as 120 and 160 respectively. Let $C : (*, \mathbb{R}) \rightarrow [0, 1]$ be a check function and $C(f^t, \epsilon)$ outputs 1 only if there exists $f^t_{i,j}$, where $i \in [0, h], j \in [0, w]$ that satisfies $f^t_{i,j} > \epsilon$. Let ϵ_T be the threshold for

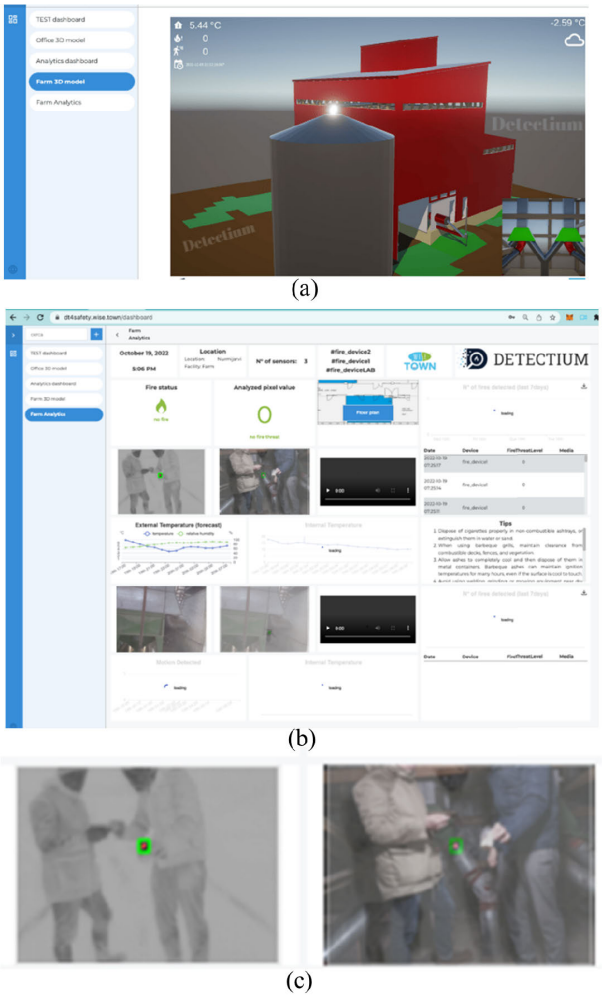


FIGURE 10. (a) The digital twin of the grain drying building with PIP of the sensor view on the cloud-based dashboard. (b) The updated dashboard integrates data feeds from both Senpack 1 and 2. (c) IR and IR-RGB overlay images showcasing the successful detection of a candle flame, uploaded to the cloud-based dashboard.

the high temperature, in our setting, $\epsilon_T = 332K(58.85°C)$. If $C(f^t, \epsilon_T) = 1$, which means there exists at least one temperature value in the frame f^t that above the threshold ϵ_T, we start the monitor phase for 1 seconds. Let $m \rightarrow (m_0, m_t)$ be the label that indicates the monitor phase, where

$$m_0 = \begin{cases} 0, & \text{if in the monitor phase} \\ 1, & \text{if not in the monitor phase} \end{cases} \quad (1)$$

m_t records the time that starts to monitor. If $C(f^t, \epsilon_T) = 1$ satisfies during the whole monitor phase, we end the monitor phase and start the variation phase. The same as the monitor phase, let $v \leftarrow (v_0, v_t)$ be the label that indicates the variation phase, where

$$v_0 = \begin{cases} 0, & \text{if in the variation phase} \\ 1, & \text{if not in the variation phase} \end{cases} \quad (2)$$

v_t records the time the variation check starts. The variation between two frames f^{t1}, f^{t2} is defined as the subtraction

of each pixel in the frame, $f^{t2} - f^{t1} : f_{ij}^{t2} - f_{ij}^{t1}$, where $i \in [0, h], j \in [0, w], t2 > t1$.The variation phase will last for 4 seconds. During the phase, we store the frame f^t into a list L every 1 seconds, when the length of the list L is at least 3, we get the current frame and the previous two frames from the list L and check the variation requirements $V(f^{t1}, f^{t2}, f^{t3}, \epsilon_V)$:

$$V(f^{t1}, f^{t2}, f^{t3}, \epsilon_V) = 1 \iff \exists i, j \begin{cases} C(f_{ij}^{t1}, \epsilon_T) = 1 \\ C(f_{ij}^{t2}, \epsilon_T) = 1 \\ C(f_{ij}^{t3}, \epsilon_T) = 1 \\ C(f_{ij}^{t2} - f_{ij}^{t1}, \epsilon_V) = 1 \\ C f_{ij}^{t3} - f_{ij}^{t2}, \epsilon_V) = 1 \end{cases} \tag{3}$$

where $i \in [0, h], j \in [0, w], t3 > t2 > t1, \epsilon_V$ is the variation threshold, in our settings, it is 5. If this satisfies, we set a fire alarm. The fire alarm consists of four tasks: upload fire alarm to cloud-based dashboard, dashboard localizes and visualizes the fire; take the picture from both the thermal camera and visual camera and generate the overlay image; record a 5 seconds video of the fire scene; sleep for 30 seconds to make sure that all the previous three tasks are finished, and clean the variation label v and go to the next round of the while loop. Notice that these four tasks will be performed in parallel, which makes the processes more efficient. If there is no fire alarm detected in the variation phase, it will stop the variation phase and go to the next round of the while loop.

Algorithm 1 outlines the primary steps for fire detection by a digital twin solution. In real-world deployment, our digital twin system also incorporates additional functionalities such as fire hazard identification, significant fire detection, sensor calibration detection to avoid false alarms, onsite buzzer alarms, mobile application notifications via a messaging platform, and object detection using computer vision algorithms.

E. SIMULATION OF HAZARDS IN REAL-TIME

The digital twin in the second case study was enhanced to allow for real-time fire simulation based on the detected fire location and size. This enables better planning for extinguishment and more effective use of access and escape routes. Various parameters were considered for creating a simple real-time fire propagation simulation, including material type, dimensions, and locations of items, which are explained in detail below.

Assuming that during regular environment monitoring, the sensor detects a fire, localizes its starting position (x_0 and y_0), and estimates its size (z_0) at time T_0, the data is then visualized in the 3D model of the digital twin in real-time. At T_0, the digital twin initiates the generation of a simple but instant fire propagation simulation model. The simulation model retrieves the existing data related to the material types, dimensions of the specific building space, and environmental objects from the digital twin's dimensionally accurate 3D model. Combining this data with temperatures collected by the sensor from the ongoing incident and known combustion rates of the material, a fire propagation simulation is rendered and uploaded as a short video to the cloud-based dashboard. The fire simulation includes the propagation of fire until the fire consumes the combustible material and is extinguished after the space is entirely burned. At T_1, the cycle is repeated,

Algorithm 1 Fire Detection Algorithm

Require: high temperature threshold ϵ_T, variation threshold ϵ_V
Ensure: Fire Alarm

```
while True do
    f^t ← output one frame from the thermal camera at time t
    if C(f^t, ε_T) = 1 then                ▷ High Temperature is Detected
        if m=(0,0) then                    ▷ Not in the monitoring phase
            if v=(0,0) then                ▷ Not in the variation phase
                m ← (1, t)                 ▷ Start to monitor and record the current time
            else                           ▷ In the variation phase
                if t − v_t ≥ 1 then
                    Add f^t to List L
                    f^t1, f^t2, f^t3 ← L if the length of L is larger than 3
                    if V(f^t1, f^t2, f^t3, ε_V) = 1 then ▷ Fire Detected
                        Do in parallel
                            Upload fire alarm to cloud-based dashboard, dashboard localizes and visualizes the fire
                            Take pictures, generate the overlay image
                            Record a 5 seconds video
                            Sleep for 30 seconds, v ← (0, 0), jump to step 2
                        end parallel
                    end if
                    if The length of L = 5 then
                        v ← (0, 0)         ▷ Stop the variation phase
                    end if
                end if
            end if
        else                               ▷ In the monitoring phase
            if t − m_t ≥ 1 then
                m ← (0, 0)                 ▷ Stop monitoring
                v ← (1, t)                 ▷ Start the variation phase
            end if
        end if
    else
        m ← (0, 0), v ← (0, 0)             ▷ High temperature is not detected
    end if
end while
```

and based on the new data from the sensor related to the location, size, and temperature of the fire, the simulation is enhanced (Fig 11).

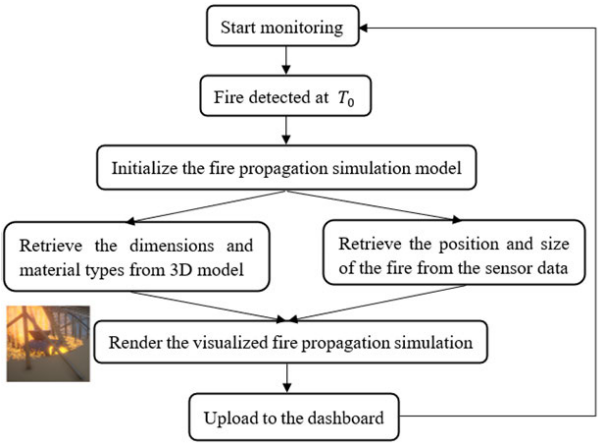


FIGURE 11. Digital twin's simulation algorithm.

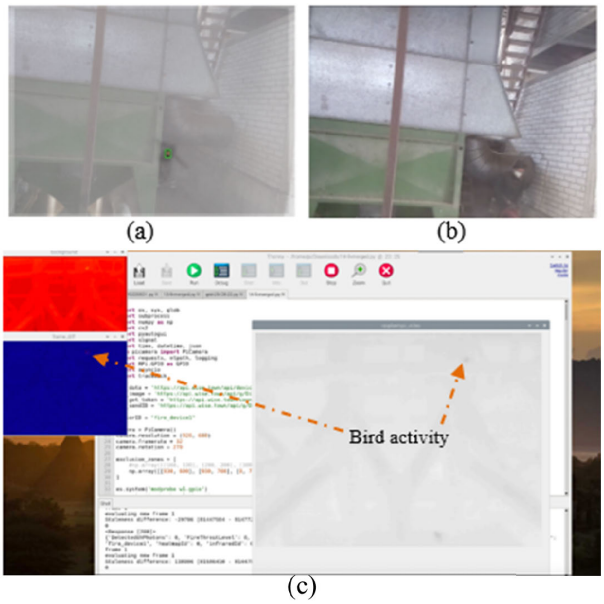


FIGURE 12. (a) A leaf blower's hot engine is detected and marked as an area of fire hazard. (b) A screenshot from the video of the event clearly shows the reason for the fire hazard detection. (c) The bird is visible as an area of temperature variation.

F. ADDITIONAL FINDINGS FROM THE CASE STUDIES

The digital twin systems were consistently tested with fire events, demonstrating high accuracy and reliability in detecting small flames. Beyond fire detection, the system also proved effective at identifying other events and anomalies. In this section, we present examples of the anomaly detection capabilities of the digital twin system.

It was observed that the digital twin can accurately monitor the uptime and downtime of the grain drying facility. For example, the operation of the hot air blower was clearly visible in the IR spectrum view. In one instance, we were able to detect the operation time span of the grain drying facility by monitoring the operational temperature of the hot air blower using data collected by the digital twin. As a result, the heat signature serves as a highly reliable indicator of machine operation, which in turn allows for conducting indirect facility/machinery utilization analysis.

In an instance demonstrating the sensitivity of the digital twin, an area of fire hazard created by significant heat from a leaf blower used to clean the area after the grain drying season was detected and registered on the cloud-based dashboard (Fig 12 (a), (b)). Such incidents, in addition to providing an opportunity to confirm the alertness of the system, can be used to improve the anomaly detection capabilities of the digital twin. This can be achieved by marking the event with computer vision and adding the machinery (in this case, the leaf blower) to the items with NOS. In the future, the alarm will only be sent to the owner/operator when the leaf blower or other machinery is operating outside its NOS temperature range.

During our second case study, we encountered an interesting event where the Senpack 2 detected the movement of birds as areas of temperature variations (Fig. 12 (c)). This is because birds and other mammals present a temperature volatility in the environment. Such events can be similar to intrusion detection and demonstrate the potential for the digital twin for safety to provide security features that are independent of motion detectors. This presents an opportunity for added features that are not possible with conventional fire detection sensors, such as smoke detectors and IR detectors. For instance, the sensor package detecting motion in the environment based on this principle can incorporate a space utilization algorithm that can help with rescue missions during and after an incident.

One of the challenges faced by every fire detection system is false positive alarms. Conventional approaches to address this issue while using a thermal camera include the use of two dimensional (2D) exclusion zones in the thermal image. Exclusion zones refer to areas considered as low fire hazard areas but can still create a significant number of false alarms due to processes or events involving high-temperature equipment and systems. However, the use of 2D exclusion zones can create inaccuracies that result in the inclusion of hazard areas in the fire detection exclusion zones.

A feature of a digital twin-enabled fire detection system is the possibility of establishing 3D exclusion zones to mitigate false alarms. Access to pixel distance data allows the system to evaluate whether the collected data belongs to the 3D fire detection exclusion zone or not. For example, in Fig 13 from the first case study, the cube on the table near the laptop is a 3D exclusion zone assigned to a low-risk area of the environment. During the monitoring of the environment, the excluded spaces are exempted from monitoring and therefore cannot trigger false fire alarms.

The case studies also helped us understand the current challenges in real-world implementation and usability of



FIGURE 13. The 3D exclusion zone illustration.

digital twins for safety in the context of buildings. One of the main challenges is confusion and a lack of understanding and adoption of novel technologies, as well as a reluctance to experiment with the new smart and revolutionary features of digital twins by building managers and farm owners. This hinders the necessary investment to facilitate the swift development of digital twin solutions for their use cases. The second challenge is related to the lack of regulations for agricultural facilities, including farms that have no obligation for fire or anomaly detection systems in their barns, and therefore rely on recuperating fire damage costs from their insurance partners. Interestingly, although cost and technological development are still among the limiting factors for implementing a digital twin, especially for the creation of manual 3D models, these obstacles are increasingly becoming a thing of the past with the availability of very powerful and economical sensors and edge computing hardware. As mentioned earlier, the manual process of generating an accurate 3D model still requires more automation and further technical development to become more useful for digital twins, especially when combined with advancing technologies such as augmented reality (AR), virtual reality (VR), and extended reality (XR).

V. DISCUSSION

While the case studies in this research primarily focused on the use phase of buildings, the potential implications of digital twins for safety and security extend across all phases of a building's lifecycle [75]. Digital twins can be utilized for fire detection, access control, surveillance, starting from the site preparation phase and continuing until the demolition phase. Table 2 summarizes the findings from the use cases conducted during this research, as well as the future opportunities for using digital twins in building lifecycle.

A. DIGITAL TWIN BENEFITS FOR BUILDING DESIGN AND PLANNING PHASE

In the pre-design, design, and planning phases, the experience of designers and architects traditionally determines the success of a design [76]. As a result, design companies with a good reputation and recognition tend to be favored. However, with a digital twin for a building, learnings from previous buildings can be transferred to designers through lessons learned not only during a specific phase but throughout all lifecycle phases. This enables a feedback loop in the design and planning process, allowing for significant knowledge transfer to designers of new buildings. This is illustrated in Fig 14, with arrows showing the transfer of knowledge from the digital twin throughout the building lifecycle to future projects.

In the specific case of safety and security digital twin, the building's safety track record and the security of its residents can lead to improvements during its renovation and can also transfer learnings to the design of future buildings that are safer and less prone to observed security incidents. Therefore, initiating the implementation of a digital twins from the design phase can not only enhance the safety and security of the current building but also contribute to the development of safer buildings in the future.

B. DIGITAL TWIN BENEFITS FOR BUILDING SITE PREPARATION AND CONSTRUCTION PHASES

Site preparation is the process of getting the construction site ready for the construction work. It includes activities such as determining off- and on-site roads for the site, preparing storage areas, and establishing engineering networks [77]. During the site preparation phase, a digital twin can be used to monitor the entry and exit routes and gates of the construction site, ensuring access control and security. It can also be used for monitoring the establishment of material storage spaces and conducting risk assessments.

Digital twins in the construction phase can aid in monitoring and increasing overall visibility of activities and operations within the construction site [19], [78]. The utilization of digital twins can improve material storage management, employee working hour monitoring, and project progress tracking [19]. Digital twins enable real-time supervision of work sites, in contrast to current practices that rely on manual labor for project progress supervision [19]. They also enhance security and safety regarding equipment, materials, and personnel, while allowing stakeholders to monitor and trace changes taking place at the construction site [78], [79]. In the event of an incident, digital twins can be used for root cause analysis. These benefits are in addition to other safety features of digital twins, such as early fire and anomaly detection, which help with downtime minimization.

C. DIGITAL TWIN BENEFITS FOR BUILDING USE AND RENOVATION PHASES

During the building use and maintenance phase, digital twins can have various applications such as improving energy efficiency, enhancing resident comfort in aspects like heating, lighting, ventilation, and automation [15]. Previous research suggests that digital twins can be beneficial for indoor safety [79]. Our research demonstrated that in terms of safety

TABLE 2. Insights and future prospects of digital twins in building lifecycle phases.

LIFECYCLE PHASES	LEARNINGS FROM CASE STUDIES	FUTURE OPPORTUNITIES
Pre-design,design,planning	Fire safety tips learned in the previous projects can be Incorporated into the new building design	Learnings from all lifecycle phases of previous projects captured in digital twins can be incorporated into various aspect of a building safety and security while developing new projects
Site preparation	Digital twin can present a method to indirectly analyze the utilization rate of machinery while also providing fire and anomaly monitoring	Security of site and safety of workers can be enhanced using digital twin for accurate and real time access control, surveillance, and risk monitoring
Construction	Actively monitoring fire risks and other anomalies, such as equipment downtime, helps to enhance work conditions and safety	Real-time monitoring, supervision, and analysis of project progress, worker safety, site security, and faster response to incidents.
Use and maintenance	Swift and accurate fire and anomaly detection, precise fire localization, fire intensity evaluation, fire propagation simulations, enhanced visibility and data sharing during an incident, data-driven insurance claim filing in the event of fire	Predictive maintenance of building assets, root cause analysis of incidents, smarter control against theft, faster and more effective reaction against fire
Renovation	Anomaly detection and fire prevention	Improved safety by active risk monitoring, access and environmental control system management according to the renovation activities in the space, also work progress monitoring
Demolition	Using digital twin's external sensors risk of fire can be monitored	Monitoring compliance with security code during demolition

and security during the building use phase, fire detection is a crucial area where digital twins can significantly contribute to early detection, fire localization, fire intensity analysis, and real-time simulation. For example, in a parking garage with electric cars, the risk of fire can be minimized through careful monitoring of the cars using a digital twin. Moreover, digital twins can enable predictive maintenance of critical building equipment, which in turn allows for improved resident comfort while also mitigating the risk of equipment failure and fire incidents. Digital twins can be used for maintaining building assets, systems, and processes [80].

In the building use phase, digital twins and their vision systems can enhance access control and security by creating a more robust smart control system to prevent theft and disturbances while also providing smoother access for building residents. This technology enables faster response in case of incidents and can also predict and prevent incidents. When an incident occurs, data sharing is quicker and more comprehensive. For post-incident root cause analysis, a digital twin can offer significant advantages.

During the renovation phase, the risk of incidents must be minimized since mitigation systems are typically offline, leading to a higher rate of incidents compared to normal building operations. Therefore, a digital twin, which should be independent of the building's operating system and have its own power source (such as a UPS), can maintain the monitoring and hazard detection processes during this high-risk period, ensuring safety and security. Renovation projects often operate under significant pressure to reduce the time residents cannot use their apartments [81]. Improved control of renovation activities using digital twins can alleviate this situation. Incidents during this phase can be costlier than those during the construction phase, as they occur in a completed building. Digital twins can significantly contribute to monitoring renovation project progress and subcontractor performance. Furthermore, similar to the construction phase, using digital twins allows for root cause analysis of potential incidents at the renovation site and overall risk and anomaly monitoring.

D. DIGITAL TWIN IN DEMOLITION PHASE

Before the start of a building's demolition process, digital twin sensors should be removed from the building's interior so that they can be used in future projects. Additional sensors can be installed outside the building and within its perimeter to monitor the demolition process. The visual sensors enable the monitoring of the process from outside the building, ensuring that security codes are followed. In case of incidents occurring, the cause can be traced back and responsible parties identified. The procedure may vary depending on the type of demolition and sensors used. The most generic process may involve removing visual sensors from the building and placing them in the surrounding area to continue monitoring from a safe distance. After the building demolition, the information collected in the digital twin's 3D model and analytics dashboard remains accessible. This data continues to accommodate the lessons learned [78], which can then be analyzed and applied to future buildings and their digital twins.

E. DIGITAL TWIN AND THE INTERPLAY BETWEEN BUILDING LIFECYCLE PHASES

Fig. 14 illustrates the building lifecycle phases and the impact of digital twins on them, as well as the fluid interaction

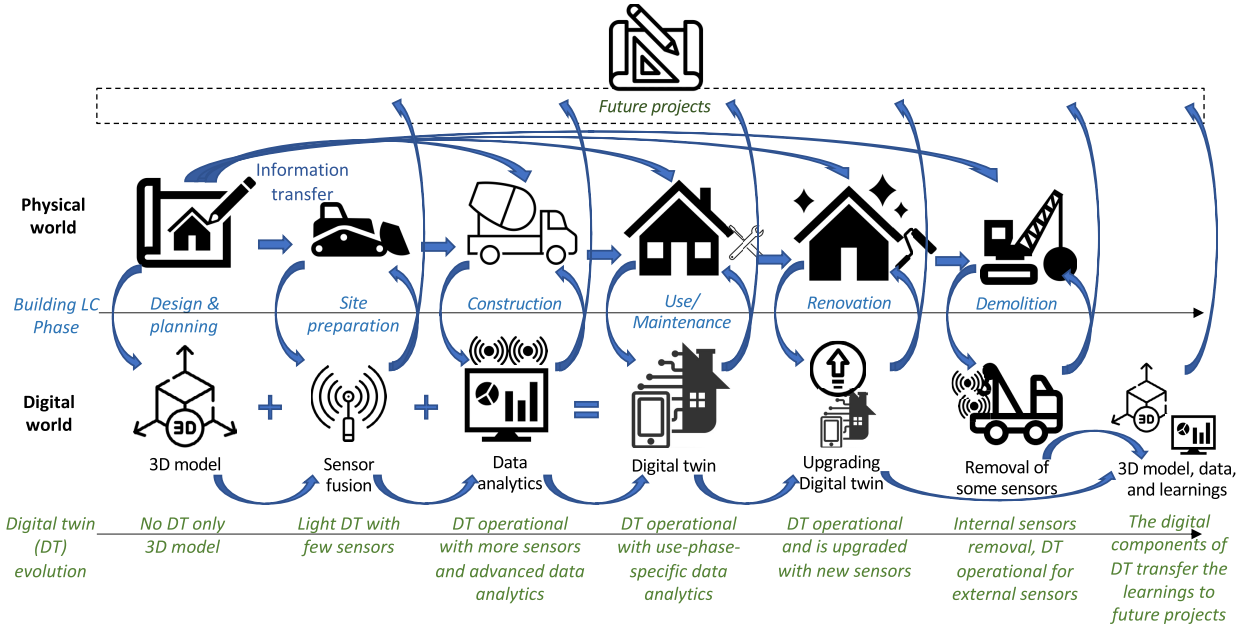


FIGURE 14. The building lifecycle phases and the digital twin impact on them, along with the dynamic interplay between these phases in both the digital and physical realms.

between these phases in digital and physical worlds. In the physical world, each phase of the lifecycle receives input from the design phase and its immediate predecessor phase. In the digital world, the digital twin is created and completed gradually, evolving from a collection of components and specific functionality during the design, site preparation, and construction phase of a building into a comprehensive and fully functional digital twin at the beginning of the use phase. This digital twin can start with specific but limited functionalities in the initial building lifecycle phases and improve to facilitate and optimize various aspects of building operations throughout the remainder of its lifecycle.

Information flow between the physical and digital world begins from the design and planning phase in the physical world. The information transferred during this phase, including the 3D design model of the building and related specifications and datasheets, is carried throughout the digital twin's evolution. Additionally, in each step of the digital twin's evolution, on top of the historical sensor data collected from the building, more data about the building's digital twin itself, including APIs, sensor types, analytic algorithms, and real-time reporting mechanisms, are added and carried forward. The digital twin's evolution occurs in parallel with the physical world's building lifecycle phases, creating an information feedback loop at each phase. This enables the benefits of using a digital twin to be realized and optimized throughout the building's lifecycle.

The digital twin carries a mix of information generated during the design phase, digital twin evolution, as well as records of events and incidents associated with each phase of the real-world building lifecycle and corresponding feedback loops. As the building reaches the end of its lifecycle, digital twin sensors may be removed, but other components, such as the 3D model and valuable insights gathered, can be harnessed in future projects to inform and elevate their development. In this way, digital twins can play a crucial role in leveraging learnings from current and past buildings for future building projects. Conventionally, knowledge gained from projects tends to stay with the people responsible for the lifecycle phases they were engaged in [82]. However, insights about similar issues from past projects can be used to address problems in the next generation of assets [83]. Digital twins enable the transfer of these learnings to future projects at any phase of the building lifecycle, fostering continuous learning and improvement.

In Fig. 14, future projects represent separate buildings, each with its distinct life-cycle. A digital twin learning interface can leverage the life-cycle twin of one building to inform the decision-making and design process of another building. Additionally, there is an "upper level" decision-making process for the focal building with a digital twin. This process encompasses determining the timeline for designing renovations, initiating risk mitigation and emergency response planning, and providing safety education for users and construction workers. These decision items pertain to both new projects and the ongoing management of the monitored building and its users [84].

F. INTERNET OF DIGITAL TWIN

Fig. 14 does not encompass an essential concept: the potential for an Internet of Digital Twins (IDT), which enables interactions between digital twins of different buildings at various stages of their life-cycles. This interconnectedness can significantly enhance the operational and optimization

aspects of entire cities, and even countries, from a broader perspective.

VI. CONCLUSION

The contribution of this paper is in the development and examination of a digital twin for fire safety, utilizing insights from two real-world case studies. We evaluate its performance and challenges, and report on the implementation and outcomes from these case studies. Furthermore, we extrapolate the learnings from the building's use phase to other lifecycle phases of buildings and facilities.

Our research approach involved several steps. We began by conducting a literature review of tools and techniques for ensuring safety and security in the context of buildings. Next, we analyzed the application of digital twins and other methods for building safety and security. Upon identifying the research gap, we undertook a significant data collection effort, followed by an analysis of the data gathered from the real-world case studies focusing on digital twin usage for safety and fire detection.

In this research, we presented a comprehensive architecture of a digital twin developed for safety, focusing on anomaly and fire detection. We highlighted its features, which include real-time data fusion, early fire and anomaly detection, fire localization, intensity evaluation, and the ability to share real-time incident data while generating fire propagation simulations based on parameters retrieved from dimensionally accurate 3D models and thermal camera sensors. We also detailed the logic behind the fire and anomaly detection algorithm and summarized the current challenges faced by digital twins for safety in the building context, including technical and customer adoption. Furthermore, we analyzed the lifecycle applications of digital twin technology for safety and security, providing a comprehensive illustration of the interplay between digital and real-world activities involved in the building lifecycle and digital twin evolution throughout the building lifecycle.

While the potential benefits of digital twins for safety and security in buildings are significant and could potentially lead to a substantial reduction in incidents and damages caused by fire and other anomalies or events, the cost and lack of knowledge about this technology are currently hindering the realization of digital twin implementation for safety and security in buildings.

A limitation of this research was the lack of access to case studies from other building lifecycle phases to validate the benefits and features discussed. Therefore, future research could utilize real-world building digital twin implementation case studies from other lifecycle phases to further study the benefits and shortcomings of digital twin technology. Additionally, since the focus of this research was on the unique features and applications of a digital twin for safety and security, future research could specifically compare the speed and accuracy of fire detection by a digital twin against various other conventional methods in a lab setting, providing quantitative results to demonstrate its potential impact. Moreover, to further enhance the extrapolations of this research on the use of digital twins for security, we suggest conducting experiments that integrate building's access control systems with the digital twin. This would complement the video surveillance system and should be considered for future work. Furthermore, as the sustainability in construction gains importance, future studies can delve into the implications of building digital twin for safety and security on environmental, social, and economic sustainability of buildings and surrounding community. Similarly, the impact of this research can also be considered in relation to Environmental, Social, and Governance (ESG) goals. For instance, from a social perspective, a digital twin for safety and security can address workplace hazards and risks, aligning with ESG goals. From an environmental standpoint, the system proposed in this article can impact air pollution and greenhouse gas emissions resulting from fire incidents, such as wildfires or industrial facility fires.

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